Code No: 136FB **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
FABRICATION PROCESSES
(Common to CE, EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define volumetric shrinkage.	[2]
b)	What is the purpose of riser in casting?	[3]
c)	Explain resistance welding.	[2]
d)	What is braze welding?	[3]
e)	What is recrystallization temperature?	[2]
f)	What is strain hardening?	[3]
g)	What fluids are used in hydrostatic extrusion?	[2]
h)	What is tube extrusion?	[3]
i)	Define open die forging.	[2]
j)	What is rolling, write its applications?	[3]

PART – B

(50 Marks)

2.a) Sketch and explain the working of a crucible furnace. Give its applications.
b) What are the elements of Gating System? Explain with sketches. [5+5]

OR

3.a) Explain various stages in investment casting process using suitable sketches. Also, list the applications of the process.
b) What are the casting defects? Explain the reasons for such defects and suggest suitable remedies. [5+5]

4.a) Explain types of welding joints with sketches. Also, give different welding positions.
b) Describe different types of gas welding flames with sketches. Suggest suitable flame type for a given metal. [5+5]

OR

5.a) Give the advantages and limitations of TIG welding over MIG welding.
b) Discuss various destructive and non-destructive methods of testing weld quality. [5+5]

6.a) What are the specific merits of cold working over hot working?

b) A cup of height 10 mm, diameter 5 mm and corner radius 0.2 mm is to be made from sheet metal of 2.5 mm thickness. Find the suitable number of draws? [5+5]

OR

7.a) The shear strength of cold rolled steel is 300 MPa. Determine the punch force required for making a blank of 120 mm diameter from a strip of 3 mm thickness of the above material. What is the theoretical and maximum punch force required?

b) Sketch and explain wire drawing and Tube drawing processes in detail. Give their applications and advantages. [5+5]

8.a) Distinguish between cold and hot extrusion processes.

b) List out the advantages, disadvantages and limitations of extrusion. [5+5]

OR

9. Explain with a neat sketch impact Extrusion process. [10]

10.a) Distinguish between drop and press forging process with reference to the process and products obtained.

b) Classify forging defects and give the necessary remedies. [5+5]

OR

11.a) Explain the rolling process in detail with the help of a sketch. What is Rolling Hill?

b) Give the advantages of closed die forging over open die forging. [5+5]

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